

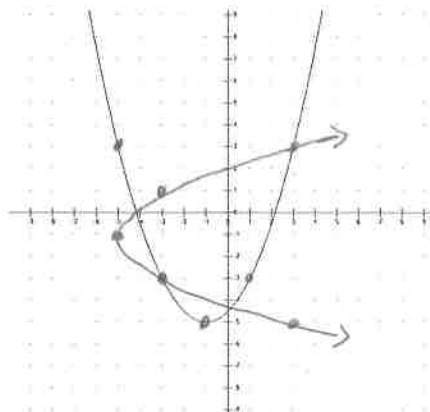
2.7 Inverse of a Function - Worksheet

MCR3U

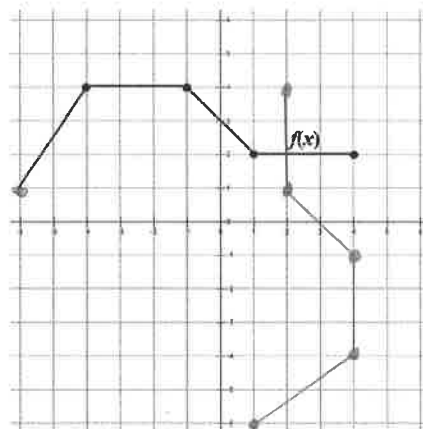
SOLUTIONS

1) Sketch the graph of the inverse of each function. Is the inverse of $f(x)$ a function? Explain.

a)



b)



2) Determine the equation of the inverse of each function.

a) $f(x) = 2x$

$$y = 2x$$

$$x = 2y$$

$$\frac{x}{2} = y$$

$$f^{-1}(x) = \frac{x}{2}$$

b) $f(x) = 6x - 5$

$$y = 6x - 5$$

$$x = 6y - 5$$

$$x + 5 = 6y$$

$$\frac{x + 5}{6} = y$$

$$f^{-1}(x) = \frac{x + 5}{6}$$

c) $f(x) = \frac{2x + 4}{5}$

$$y = \frac{2x + 4}{5}$$

$$x = \frac{2y + 4}{5}$$

$$5x = 2y + 4$$

$$5x - 4 = 2y$$

$$\frac{5x - 4}{2} = y$$

$$f^{-1}(x) = \frac{5x - 4}{2}$$

3) Determine the equation of the inverse of each function

a) $f(x) = x^2 + 6$

$$y = x^2 + 6$$

$$x = y^2 + 6$$

$$x - 6 = y^2$$

$$\pm\sqrt{x - 6} = y$$

$$f^{-1}(x) = \pm\sqrt{x - 6}$$

b) $f(x) = (x + 8)^2$

$$y = (x + 8)^2$$

$$x = (y + 8)^2$$

$$\pm\sqrt{x} = y + 8$$

$$\pm\sqrt{x} - 8 = y$$

$$f^{-1}(x) = \pm\sqrt{x} - 8$$

4) For each quadratic function, complete the square and then determine the equation of the inverse.

a) $f(x) = x^2 + 6x + 15$

$$f(x) = (x^2 + 6x + 9 - 9) + 15$$

$$f(x) = (x+3)^2 + 6$$

$$x = (y+3)^2 + 6$$

$$x-6 = (y+3)^2$$

$$\pm\sqrt{x-6} - 3 = y$$

$$f^{-1}(x) = \pm\sqrt{x-6} - 3$$

b) $f(x) = 2x^2 + 24x - 3$

$$f(x) = 2(x^2 + 12x + 36) - 72 - 3$$

$$f(x) = 2(x+6)^2 - 75$$

$$x = 2(y+6)^2 - 75$$

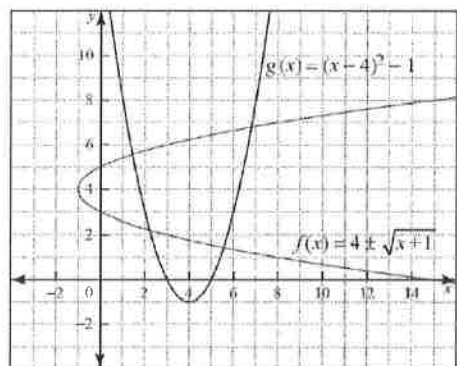
$$\frac{x+75}{2} = (y+6)^2$$

$$\pm\sqrt{\frac{x+75}{2}} - 6 = y$$

$$f^{-1}(x) = \pm\sqrt{\frac{x+75}{2}} - 6$$

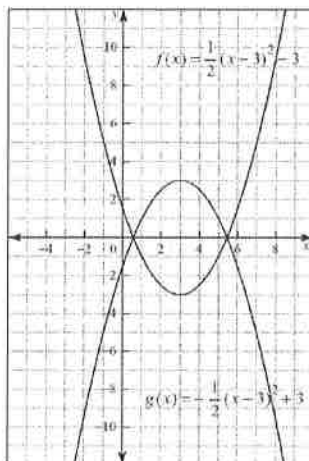
5) Determine if the two relations shown are inverses of each other. Justify your conclusion.

a)



yes

b)



NO

6) For the function $f(x) = -5x + 6$

a) determine $f^{-1}(x)$

b) Graph $f(x)$ and its inverse

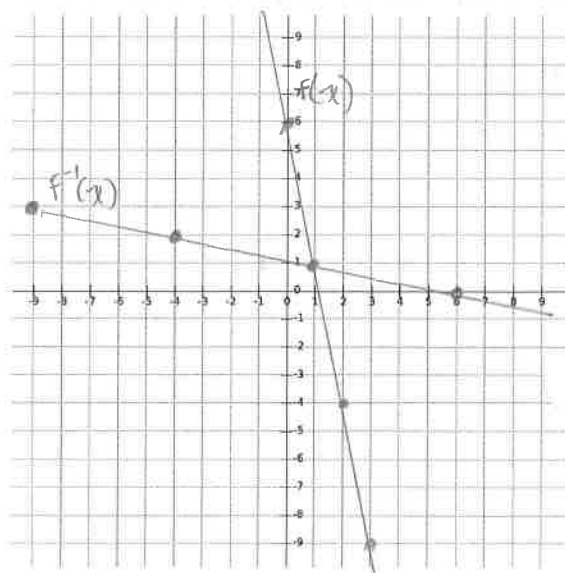
$$x = -5y + 6$$

$$x-6 = -5y$$

$$\frac{x-6}{-5} = y$$

$$-\frac{x-6}{5} = y$$

$$f^{-1}(x) = \frac{-x+6}{5}$$



7) Use transformations to graph the function $f(x) = 2(x - 2)^2 + 1$. Find the inverse function $f^{-1}(x)$ and graph it by reflecting $f(x)$ over the line $y = x$ (switch x and y co-ordinates)

- 1) vertical stretch to 2 ($2y$)
- 2) shift right 2 units ($x+2$)
- 3) shift up 1 unit ($y+1$)

$$y = x^2$$

$$(-3, 9)$$

$$(-2, 4)$$

$$(-1, 1)$$

$$(0, 0)$$

$$(1, 1)$$

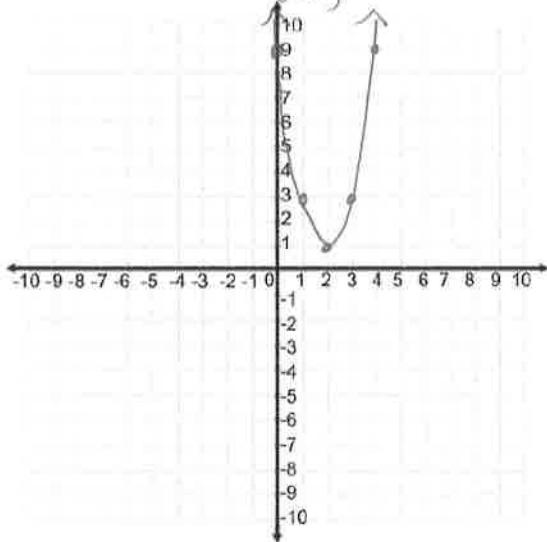
$$(2, 4)$$

$$(3, 9)$$

$x+2$	$2y+1$
-1	19
0	9
1	3
2	1
3	3
4	9
5	19

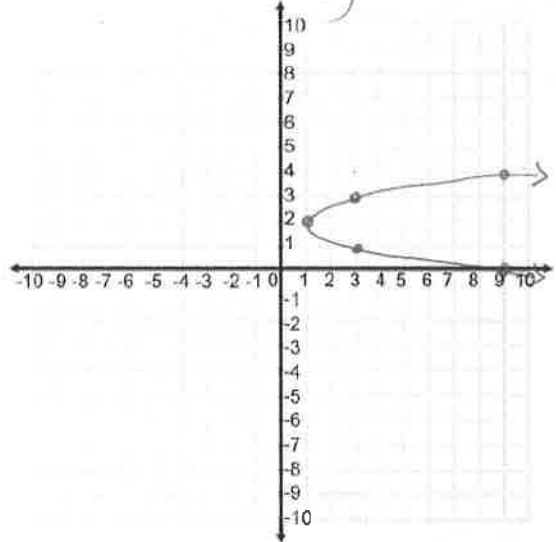
graph these

$f(x)$



x	y
19	-1
9	0
3	1
1	2
3	3
9	4
19	5

$f^{-1}(x)$



8) Determine the equation of the inverse for the given functions and state the domain and range.

a) $f(x) = \sqrt{x+3}$

$D: \{x \in \mathbb{R} \mid x \geq -3\}$

$R: \{y \in \mathbb{R} \mid y \geq 0\}$

$$x = \sqrt{y+3}$$

$$x^2 = y+3$$

$$x^2 - 3 = y$$

$$f^{-1}(x) = x^2 - 3$$

$D: \{x \in \mathbb{R} \mid x \geq 0\}$

$R: \{y \in \mathbb{R} \mid y \geq -3\}$

b) $f(x) = \frac{3}{x-2} + 2$

$D: \{x \in \mathbb{R} \mid x \neq 2\}$

$R: \{y \in \mathbb{R} \mid y \neq 2\}$

$$x = \frac{3}{y-2} + 2$$

$$x-2 = \frac{3}{y-2}$$

$$y-2 = \frac{3}{x-2}$$

$$y = \frac{3}{x-2} + 2$$

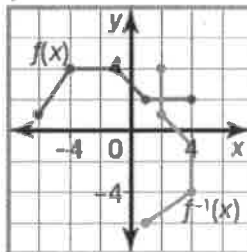
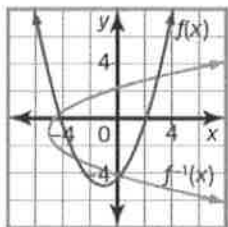
$$f^{-1}(x) = \frac{3}{x-2} + 2$$

$D: \{x \in \mathbb{R} \mid x \neq 2\}$

$R: \{y \in \mathbb{R} \mid y \neq 2\}$

Answers

1) a) the inverse is NOT a function b)



inverse is NOT a function

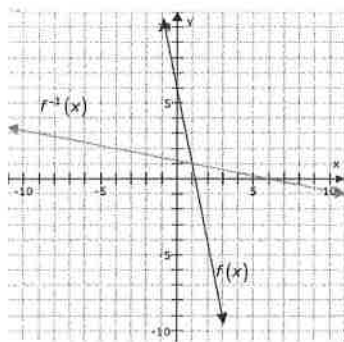
2) a) $f^{-1}(x) = \frac{x}{2}$ b) $f^{-1}(x) = \frac{x+5}{6}$ c) $f^{-1}(x) = \frac{5x-4}{2}$

3) a) $f^{-1}(x) = \pm\sqrt{x-6}$ b) $f^{-1}(x) = \pm\sqrt{x} - 8$

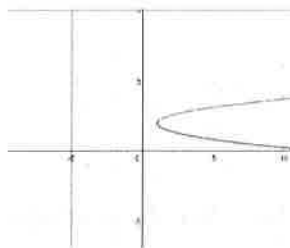
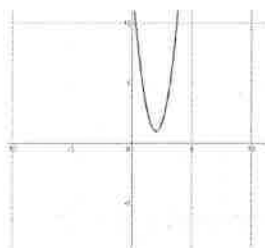
4) a) $f^{-1}(x) = \pm\sqrt{x-6} - 3$ b) $f^{-1}(x) = \pm\sqrt{\frac{x+75}{2}} - 6$

5) a) yes b) no

6) a) $f^{-1}(x) = \frac{-x+6}{5}$ b)



7) $f^{-1}(x) = 2 \pm \sqrt{\frac{x-1}{3}}$



8) a) $f^{-1}(x) = x^2 - 3$; Domain for $f(x)$: $\{X \in \mathbb{R} | x \geq -3\}$, Range for $f(x)$: $\{Y \in \mathbb{R} | y \geq 0\}$
Domain for $f^{-1}(x)$: $\{X \in \mathbb{R} | x \geq 0\}$, Range for $f(x)$: $\{Y \in \mathbb{R} | y \geq -3\}$

b) $f^{-1}(x) = \frac{3}{x-2} + 2$; Domain for $f(x)$: $\{X \in \mathbb{R} | x \neq 2\}$, Range for $f(x)$: $\{Y \in \mathbb{R} | y \neq 2\}$
Domain for $f^{-1}(x)$: $\{X \in \mathbb{R} | x \neq 2\}$, Range for $f(x)$: $\{Y \in \mathbb{R} | y \neq 2\}$